

MAINTENANCE WORK ORDER

Work Order No. : WO-4892
Date Issued : 22 January 2026
Priority : CRITICAL
Status : IN PROGRESS
Department : Instrumentation & Electrical / HSE
Location : Bay 4, Zone C
Equipment Tag : GB-14 (Draeger Polytron 2 XP Combustible Gas Detector)
Assigned To : Anil Mehta, Senior Technician (I&E)
Supervised By : Rajiv Sharma, Maintenance Supervisor
HSE Sign-off : Kavita Desai, HSE Manager (replacing Priya Nair, retired Dec 2025)
Approved By : Deepak Singh, Maintenance Engineer

1. WORK DESCRIPTION

URGENT: Suspension and investigation of gas sensor GB-14 following sustained elevated reading of 22% LEL recorded during Night Shift on 21 January 2026 (Shift Handover Log SHL-2026-0121).

A Hot Work Permit (HWP-2026-018) was active in Bay 4, Zone C on 21 January 2026 for plasma-cutting operations on pressure vessel PV-08 structural support bracket. The permit was issued at 21:30 IST by Rajiv Sharma. At 23:15 IST, Operator Suresh Kumar (Night Shift) noted GB-14 reading 22% LEL and suspended hot work.

THIS IS THE THIRD OCCURRENCE of elevated GB-14 readings coinciding with active hot work in Bay 4. Previous occurrences:

- 1. March 2019 ? 27% LEL, near-miss, NMR-2019-047
- 2. August 2024 ? 18% LEL, sensor drift + HX-11 flange seepage, WO-4471

Pattern noted by Rajiv Sharma. Immediate escalation to Plant Manager Mohan Rao and HSE Manager Kavita Desai at 23:45 IST on 21 January 2026.

2. IMMEDIATE ACTIONS (22 JAN 2026)

- 01:00 ? All hot work in Bay 4 suspended indefinitely pending investigation.
- 02:30 ? Portable multi-gas detector deployed by Anil Mehta. Reading at GB-14 location: CH4: 19% LEL, CO: 12 ppm, H2S: 0 ppm. Methane source confirmed present.
- 05:00 ? HX-11 flange joint #7 (same joint flagged in WO-4476, Aug 2024) found to have progressive seepage ? compression gasket degraded. Emergency isolation actioned.
- 06:15 ? Bay 4 ventilated. Methane at GB-14 location: 2% LEL. Area safe.
- 07:00 ? GB-14 calibration checked. Zero drift: +3.8% LEL. Span: 48% LEL at 50% reference. Sensor out of tolerance. Emergency recalibration performed.

3. ROOT CAUSE INVESTIGATION (in progress)

Preliminary root causes identified (formal RCA to follow under RCA-2026-007):

- (a) HX-11 flange joint #7: Gasket (spiral-wound, 150# ANSI) degraded due to thermal cycling. This is the SAME joint identified in WO-4476 (August 2024) where seepage was noted. WO-4476 was closed as "monitoring required" without gasket replacement. Corrective action from WO-4476 was not completed.
- (b) GB-14 sensor element: Last replaced per WO-4473 (November 2024). Replacement was overdue 6 months prior (first noted in WO-4471). At 14 months of use, sensor is within manufacturer's 24-month replacement interval but calibration drift of +3.8% LEL suggests early degradation ? possibly due to Bay 4 methane exposure.
- (c) Hot Work Permit HWP-2026-018: Issued by Rajiv Sharma without live GB-14 reading check. Procedure updated in 2019 (CA-02 from NMR-2019-047) requires live reading within 30 minutes. Permit paperwork shows the gas reading field was marked "N/A ? routine area" rather than recording an actual reading. This procedural shortcut has recurred.
- (d) Pattern recognition failure: The recurrence of this exact scenario (elevated GB-14 +